Supplementary Information for

A Paired, Floor-Guarded Evaluation Protocol for TensorRT INT8 and FP8 Object Detectors under Synthetic Image Corruptions

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S1 Purpose, scope, and evidence hierarchy

This supplement documents the estimands, diagnostic summaries, sensitivity analyses, and artifact contract supporting the main article. It does not add a new population-level claim. The evidence hierarchy used throughout the paper is: (i) the 144-cell four-dataset YOLO landscape for exploratory factorial characterization; (ii) the untouched VOC/KITTI rerun as the strongest split-independent conditional evidence; (iii) targeted realization, training, calibration, class-universe, codec, weighting, and runtime sensitivities; and (iv) RT-DETR and RetinaNet experiments as recipe-portability stress cases.

The labels *INT8* and *FP8* denote the complete recorded TensorRT treatments—quantizer placement, Q/DQ coverage, calibration or scale construction, graph rewriting, mixed-precision fallbacks, engine building, and frozen decoding—rather than datatype effects in isolation. All quantities are reported in AP points unless stated otherwise.

S2 Atomic paired design and estimands

For a fixed dataset d, checkpoint m, endpoint b, and corruption condition c, let 95 denote the deterministic JPEG-quality-95 matched-clean control. For quantized treatment q, define the FP32-referenced quantization gap and the clean-adjusted excess gap as

$$Q_{d,m,q,b,c} = \mathrm{AP}_{d,m,\mathrm{FP32},b,c} - \mathrm{AP}_{d,m,q,b,c}, \tag{S1}$$

$$E_{d,m,q,b,c} = Q_{d,m,q,b,c} - Q_{d,m,q,b,95}. \tag{S2}$$

A positive E means that corruption widens the treatment–FP32 discrepancy relative to matched clean; a negative E means that the discrepancy contracts. Neither sign is, by itself, evidence of retained absolute accuracy.

The direct treatment interaction is computed inside each common image resample:

$$\Delta E_{d,m,b,c} = E_{d,m,\mathrm{INT8},b,c} - E_{d,m,\mathrm{FP8},b,c} \tag{S3}$$

$$= \big(\mathrm{AP}_{\mathrm{FP8},c} - \mathrm{AP}_{\mathrm{INT8},c}\big) - \big(\mathrm{AP}_{\mathrm{FP8},95} - \mathrm{AP}_{\mathrm{INT8},95}\big). \tag{S4}$$

The FP32 term cancels algebraically. Consequently, ΔE asks whether the FP8–INT8 discrepancy changes under corruption after removing the discrepancy already present on matched clean inputs. A positive value indicates expansion; a negative value indicates contraction.

For area endpoints, the within-treatment size contrast is

$$\Psi_{d,m,q,c} = E_{d,m,q,S,c} - E_{d,m,q,L,c}, \tag{S5}$$

with direct size interaction $\Delta\Psi = \Psi_{\mathrm{INT8}} - \Psi_{\mathrm{FP8}}$. TT100K uses its native height-based small-like and large-like endpoints, which are reported separately and never pooled numerically with the COCO/VOC/KITTI area endpoints.

The absolute-accuracy guardrail is

$$D_{d,m,p,b,c} = \mathrm{AP}_{d,m,p,b,95} - \mathrm{AP}_{d,m,p,b,c}. \tag{S6}$$

The protocol is therefore *floor-guarded*: D, corrupted AP, and low-AP inventories are reported beside relative interactions. It is not a formal statistical correction for censoring or an AP floor.

The image is the resampling unit. Each direct comparison reuses one deterministic schedule of 2,000 image-index bootstrap draws across all four INT8/FP8 by clean/corrupted arms. Percentile intervals quantify finite-evaluation-set uncertainty conditional on the frozen datasets, checkpoints, calibration lists, engines, corruption materializations, and decoder. They do not propagate a population distribution over datasets, training runs, calibration runs, or corruption generators.

S3 Decision-impact diagnostic

The methodological value of clean adjustment can be measured directly. For each of the 144 exploratory cells, the raw corrupted gap satisfies

$$(\mathrm{AP}_{\mathrm{FP8}} - \mathrm{AP}_{\mathrm{INT8}})_c = (\mathrm{AP}_{\mathrm{FP8}} - \mathrm{AP}_{\mathrm{INT8}})_{95} + \Delta E. \tag{S7}$$

The raw corrupted comparison favored FP8 in 134 cells, but 56 of those cells became negative after subtracting the matched-clean discrepancy (Table S1). Thus a raw corrupted leaderboard comparison would conflate clean fidelity with corruption-specific interaction in 38.9% of the finite grid.

Table S1: Decision-impact inventory over the 144 exploratory cells. "Below AP" counts a cell when at least one quantized treatment crosses the stated floor.

Diagnostic	Cells
Raw corrupted FP8–INT8 gap > 0	134/144
Raw corrupted FP8–INT8 gap < 0	10/144
Clean-adjusted $\Delta E > 0$	78/144
Clean-adjusted $\Delta E < 0$	66/144
Raw-positive but adjusted-negative	56/144
At least one format below 10 AP	18/144
At least one format below 5 AP	5/144

S4 Quantized graph coverage

Table S2 summarizes hash-bound ONNX graph coverage. These counts characterize inserted Q/DQ structure and zero-point datatype; they are not a claim that every TensorRT kernel executes at the nominal precision. That stronger claim would require per-layer engine inspection or profiling, which is outside the recorded evidence package.

Table S2: Graph-level Q/DQ coverage across the evaluated detector families. Ranges reflect the three dataset-specific graphs for each row.

Family	Model	Format	Graphs	Q	Weight Q	Act. Q	Direct Q/DQ nodes	Zero point
YOLO11	YOLO11m	INT8	3	437–441	113	324–328	352–355/405	INT8
YOLO11	YOLO11m	FP8	3	213–219	101–104	112–115	133–136/405	FLOAT8E4M3FN
YOLO11	YOLO11n	INT8	3	325–341	85–88	240–253	263–275/320	INT8
YOLO11	YOLO11n	FP8	3	152–158	69–72	83–86	99–102/320	FLOAT8E4M3FN
YOLO11	YOLO11x	INT8	3	682–690	174	508–516	546–552/617	INT8
YOLO11	YOLO11x	FP8	3	346–352	161–164	185–188	215–218/617	FLOAT8E4M3FN
RT-DETR	RT-DETR-L	INT8	3	534–538	197–199	335–339	351–353/963	INT8
RT-DETR	RT-DETR-L	FP8	3	398–401	167–169	231–232	251–253/963	FLOAT8E4M3FN
RetinaNet	R50-FPN-v2	INT8	3	217	111	106	139/1076	INT8
RetinaNet	R50-FPN-v2	FP8	3	214	110	104	137/1076	FLOAT8E4M3FN

S5 Exploratory interaction summaries

Table S3 reports the four dataset margins, while Table S4 reports capacity, corruption, and ordinal-severity margins. Cell-wise percentile signs are descriptive inventories rather than multiplicity-adjusted discoveries. The balanced macro is the arithmetic mean of the finite, prespecified 144-cell grid; it is not a deployment-population weighted effect.

Table S3: Dataset-level direct interaction summaries. TT100K's $\Delta\Psi$ uses the height endpoint; the other rows use the area endpoint.

Dataset	Cells	Mean ΔE (AP pt)	95% percentile sign (+/-/cross)	Mean $\Delta\Psi$ (AP pt)
COCO	36	-0.19	6 / 10 / 20	-0.12
VOC	36	-0.04	12 / 5 / 19	-0.44
KITTI	36	-0.46	5 / 11 / 20	-0.95
TT100K	36	+0.61	2 / 0 / 34	-1.38

Table S4: Factor-level heterogeneity in the exploratory direct ledger.

Factor	Level	Cells	Mean ΔE	Range ΔE	95% percentile sign (+/-/cross)
Checkpoint rung	YOLO11n	48	+0.07	-2.71 to +3.10	11 / 7 / 30
Checkpoint rung	YOLO11m	48	-0.11	-4.28 to +2.82	9 / 10 / 29
Checkpoint rung	YOLO11x	48	-0.02	-4.23 to +2.09	5 / 9 / 34
Corruption	Gaussian noise	36	-0.03	-3.66 to +2.82	9 / 11 / 16
Corruption	Motion blur	36	-0.24	-4.28 to +1.48	4 / 8 / 24
Corruption	Fog	36	+0.39	-0.56 to +3.10	6 / 0 / 30
Corruption	JPEG	36	-0.20	-4.23 to +2.01	6 / 7 / 23
Severity	1	48	+0.28	-1.03 to +2.82	8 / 2 / 38
Severity	3	48	+0.16	-3.37 to +2.09	10 / 8 / 30
Severity	5	48	-0.51	-4.28 to +3.10	7 / 16 / 25

S6 Size-specific absolute-accuracy guardrail

Table S5 binds relative size interactions to absolute matched-clean AP, corrupted AP, and clean-to-corrupted loss. The AP$<$ 5 inventory makes explicit when an apparent contraction occurs close to a task failure floor. COCO, VOC, and KITTI use COCO-style area endpoints; TT100K uses its original-box-height endpoints.

Table S5: Size-specific absolute-accuracy guardrail. Values are balanced means across 36 model–corruption–severity cells per row.

Dataset	Format	Endpoint	Cells	Clean AP	Corrupted AP	Mean D (AP pt)	AP¡5
COCO	INT8	coco-area:small	36	28.05	18.52	+9.53	5
COCO	INT8	coco-area:large	36	61.75	49.36	+12.39	0
COCO	FP8	coco-area:small	36	29.01	19.37	+9.64	4
COCO	FP8	coco-area:large	36	63.33	50.94	+12.38	0
KITTI	INT8	coco-area:small	36	50.90	34.18	+16.72	6
KITTI	INT8	coco-area:large	36	74.89	54.12	+20.77	2
KITTI	FP8	coco-area:small	36	54.02	36.52	+17.50	6
KITTI	FP8	coco-area:large	36	76.25	55.64	+20.61	2
TT100K	INT8	original-height:XS	36	12.02	6.42	+5.60	20
TT100K	INT8	original-height:S	36	38.03	25.46	+12.57	8
TT100K	INT8	original-height:L	36	58.50	43.67	+14.83	1
TT100K	INT8	original-height:XL	36	65.22	51.82	+13.40	0
TT100K	FP8	original-height:XS	36	14.00	6.93	+7.06	21
TT100K	FP8	original-height:S	36	40.15	27.11	+13.04	8
TT100K	FP8	original-height:L	36	58.52	44.97	+13.55	0
TT100K	FP8	original-height:XL	36	66.42	53.50	+12.93	0
VOC	INT8	coco-area:small	36	28.12	19.98	+8.14	6
VOC	INT8	coco-area:large	36	71.96	54.99	+16.97	0
VOC	FP8	coco-area:small	36	28.97	20.51	+8.46	6
VOC	FP8	coco-area:large	36	72.97	56.12	+16.85	0

S7 Codec-control sensitivity

The primary estimand uses deterministic JPEG-95 matched-clean inputs so that both clean and corrupted arms share the terminal materialization codec. Table S6 compares original-source clean AP with JPEG-95 clean AP. This point sensitivity quantifies the control choice; it is not a bootstrap interval and does not replace the codec-matched estimand.

Table S6: Original-source clean versus deterministic JPEG-95 clean AP. The last column is JPEG-95 minus original-source AP.

Dataset	Model	Precision	Original clean AP (%)	JPEG-95 clean AP (%)	Codec minus original (AP pt)
COCO	YOLO11n	FP32	38.73	38.70	-0.035
COCO	YOLO11n	INT8	36.64	36.65	+0.014
COCO	YOLO11n	FP8	37.74	37.81	+0.072
COCO	YOLO11m	FP32	50.68	50.57	-0.113
COCO	YOLO11m	INT8	48.85	48.82	-0.025
COCO	YOLO11m	FP8	50.08	50.05	-0.028
COCO	YOLO11x	FP32	53.81	53.69	-0.118
COCO	YOLO11x	INT8	52.75	52.56	-0.197
COCO	YOLO11x	FP8	53.52	53.47	-0.056
VOC	YOLO11n	FP32	62.54	62.56	+0.013
VOC	YOLO11n	INT8	61.42	61.51	+0.090
VOC	YOLO11n	FP8	62.06	62.19	+0.125
VOC	YOLO11m	FP32	68.53	68.56	+0.036
VOC	YOLO11m	INT8	66.75	66.66	-0.089
VOC	YOLO11m	FP8	68.31	68.36	+0.044
VOC	YOLO11x	FP32	70.09	70.05	-0.032
VOC	YOLO11x	INT8	69.07	69.03	-0.047
VOC	YOLO11x	FP8	70.00	69.94	-0.060
KITTI	YOLO11n	FP32	64.79	64.84	+0.049
KITTI	YOLO11n	INT8	61.52	61.69	+0.170
KITTI	YOLO11n	FP8	63.68	63.83	+0.147
KITTI	YOLO11m	FP32	73.82	73.87	+0.049
KITTI	YOLO11m	INT8	69.22	69.34	+0.124
KITTI	YOLO11m	FP8	73.32	73.31	-0.016
KITTI	YOLO11x	FP32	72.93	73.06	+0.127
KITTI	YOLO11x	INT8	70.87	70.82	-0.053
KITTI	YOLO11x	FP8	72.20	72.26	+0.061
TT100K	YOLO11n	FP32	28.78	28.76	-0.013
TT100K	YOLO11n	INT8	25.10	25.29	+0.191
TT100K	YOLO11n	FP8	27.12	26.41	-0.709
TT100K	YOLO11m	FP32	56.57	56.55	-0.014
TT100K	YOLO11m	INT8	53.05	52.97	-0.074
TT100K	YOLO11m	FP8	54.75	54.40	-0.350
TT100K	YOLO11x	FP32	57.63	57.66	+0.034
TT100K	YOLO11x	INT8	55.50	56.02	+0.521
TT100K	YOLO11x	FP8	56.76	56.10	-0.667

S8 Weighting, omission, and runtime sensitivities

Alternative weighting changes the descriptive target, not merely the standard error. Table S7 therefore reports equal-cell, image-count, and object-count means together with leave-one-dataset and leave-one-corruption summaries. Sign changes near zero expose conditional heterogeneity; they do not identify a causal role for the omitted factor.

Table S7: Weighting and leave-one-factor-out sensitivity of the 144-cell ledger.

Analysis	Level	Cells	Mean ΔE (AP pt)	SD / IQR (AP pt)
Cell distribution	all cells	144	-0.02	1.20 / [-0.39, +0.56]
Weighting	equal cell	144	-0.02	–
Weighting	image weighted	144	+0.00	–
Weighting	object weighted	144	-0.09	–
Leave-one-dataset-out	COCO	108	+0.03	–
Leave-one-dataset-out	KITTI	108	+0.13	–
Leave-one-dataset-out	TT100K	108	-0.23	–
Leave-one-dataset-out	VOC	108	-0.01	–
Leave-one-corruption-out	Fog	108	-0.16	–
Leave-one-corruption-out	Gaussian noise	108	-0.02	–
Leave-one-corruption-out	JPEG	108	+0.04	–
Leave-one-corruption-out	Motion blur	108	+0.05	–

Runtime is reported only for matched engine conditions. Ratios exclude image decoding, preprocessing, host–device transfer, detector decoding, NMS, power, and energy. Table S8 stratifies the engine-only measurements by input size and YOLO11 capacity rather than pooling unlike workloads into one deployment claim.

Table S8: Matched engine-only latency-ratio sensitivity. Entries are median [Q1, Q3] across retained conditions.

Stratum	Level	Conditions	FP32/INT8	FP32/FP8	INT8/FP8
Input size	640	9	3.08 [1.56, 3.08]	3.32 [1.81, 3.33]	1.09 [1.08, 1.16]
Input size	1280	3	2.81 [2.38, 2.84]	2.86 [2.33, 2.91]	1.02 [0.97, 1.02]
Capacity	YOLO11m	4	3.08 [3.01, 3.09]	3.34 [3.21, 3.35]	1.08 [1.06, 1.09]
Capacity	YOLO11n	4	1.55 [1.52, 1.65]	1.80 [1.79, 1.81]	1.17 [1.10, 1.18]
Capacity	YOLO11x	4	3.06 [3.00, 3.09]	3.32 [3.22, 3.33]	1.07 [1.05, 1.08]

S9 Untouched-holdout and corruption-realization sensitivities

The untouched branch froze disjoint checkpoint-selection and final image lists before training. VOC used 2,510 selection and 5,823 final images; KITTI used 1,496 selection and 1,197 final images. Calibration identities were drawn only from training data. The final factorial rerun retained 72 equal-weight VOC/KITTI cells and reused a common 2,000-draw schedule within each dataset. Its balanced direct interaction was -0.55 AP points with percentile interval $[-0.78, -0.29]$; VOC was -0.43 $[-0.54, -0.29]$ and KITTI was -0.67 $[-1.13, -0.20]$. FP8 retained 1.60 more matched-clean AP on average, and the clean advantage was positive in all six dataset–capacity blocks. Nine of 144 corrupted format arms were below 5 AP and 18 were below 10 AP. This branch removes evaluation-after-selection reuse for these two datasets, but it does not convert the broader exploratory grid into a prospective confirmatory study.

The corruption-realization sensitivity used three fixed realization seeds over 36 base conditions on class-covering 512-image subsets of VOC, KITTI, and TT100K. Its equal-cell mean ΔE was -0.23 AP points, with nested realization/image percentiles $[-0.95, -0.28, +0.37]$. Mean between-realization variance was 0.3989 AP-points^2, compared with 0.7446 within-image-bootstrap variance. For $\Delta\Psi$, the mean was -1.04 with nested percentiles $[-3.05, -1.42, +0.10]$. Only three materializations were used, so these are conditional sensitivity summaries rather than population intervals over all possible corruptions. In this separate sensitivity branch, stochastic fields and motion-blur angles were nested across severity; in the exploratory grid, severity remains an ordinal factor conditional on its recorded materialization.

S10 Training–calibration seed sensitivity

The targeted 3×3 training–calibration grid covers YOLO11m on VOC, KITTI, and TT100K at severity 5. Across 108 cells, the equally weighted mean ΔE was -1.11 AP points. Training-seed marginal means ranged from -1.64 to

-0.68, and calibration-seed marginal means from -1.32 to -0.84. The original-seed intersection was -1.21, close to the balanced seed-grid mean. These results do not extend to COCO, other capacity rungs, lower severities, or an unspecified population of training and calibration seeds.

Table S9: Descriptive two-way seed decomposition within each targeted YOLO11m severity-5 block. The last three columns are percentages of the within-block total sum of squares; Train×cal. denotes non-additivity.

Dataset	Corruption	Mean	Train (%)	Cal. (%)	Train×cal. (%)
VOC	Gaussian noise	-1.494	28.6	23.0	48.4
VOC	Motion blur	-1.256	11.3	18.0	70.7
VOC	Fog	+0.018	34.2	36.5	29.3
VOC	JPEG	-2.033	9.3	80.2	10.5
KITTI	Gaussian noise	-3.532	51.5	13.8	34.8
KITTI	Motion blur	-4.350	57.1	37.4	5.5
KITTI	Fog	+1.608	23.9	24.8	51.3
KITTI	JPEG	+0.120	38.4	54.6	7.0
TT100K	Gaussian noise	-1.381	8.7	22.5	68.8
TT100K	Motion blur	-0.998	12.6	5.5	82.0
TT100K	Fog	+0.769	11.5	27.5	61.0
TT100K	JPEG	-0.787	90.9	0.5	8.5

S11 Fixed-class-universe bootstrap sensitivity

For the prespecified TT100K YOLO11m/motion-blur/severity-5 stress cell, a positive-weight Bayesian image bootstrap retained the complete category universe across all 3,067 images. With 10,000 draws, the ΔE percentiles were $[-1.19, -0.17, +0.88]$, compared with $[-1.79, -0.20, +1.34]$ under the ordinary 2,000-draw image bootstrap. The height-based $\Delta\Psi$ percentiles were $[-5.69, -4.08, -2.48]$ versus $[-7.22, -4.48, -1.81]$. The median sign and zero-crossing status of ΔE, and the negative-interval status of $\Delta\Psi$, were unchanged. This targeted cell does not establish class-composition invariance for the full TT100K grid.

S12 Architecture-portability stress cases

The RT-DETR-L and RetinaNet-R50-FPN-v2 extension tests whether the frozen quantization recipes transfer beyond the YOLO family. It is interpreted in a fixed sequence: matched-clean fidelity first, corruption interaction second, and absolute corrupted AP third.

Table S10: Matched JPEG-95 clean AP and clean quantization gaps for the architecture-portability stress cases.

Dataset	Family	FP32 AP	INT8 AP	FP8 AP	Q_{95}^{INT8}	Q_{95}^{FP8}
VOC	RT-DETR-L	70.44	1.14	70.14	69.30	0.30
VOC	RetinaNet-R50-FPN-v2	57.22	33.69	56.17	23.53	1.05
KITTI	RT-DETR-L	12.29	3.32	12.02	8.97	0.27
KITTI	RetinaNet-R50-FPN-v2	50.43	7.13	49.27	43.30	1.15
TT100K	RT-DETR-L	43.17	5.46	42.13	37.71	1.04
TT100K	RetinaNet-R50-FPN-v2	3.37	2.60	3.38	0.77	-0.01

Table S11: Canonical cross-family interaction summaries under $E = Q_c - Q_{95}$. Negative INT8 means largely reflect contraction of a large clean gap near low corrupted AP.

Family	Format	Mean Q_{95}	Mean E	E range	Mean corrupted AP	AP<5
RT-DETR-L	INT8	38.66	-6.81	-54.39–+1.22	1.97	34/36
RT-DETR-L	FP8	0.54	-0.02	-1.06–+2.01	33.30	4/36
RetinaNet-R50-FPN-v2	INT8	22.53	-6.21	-38.86–+1.14	11.34	16/36
RetinaNet-R50-FPN-v2	FP8	0.73	+0.01	-1.11–+0.64	26.91	12/36

Table S12: Descriptive direct ΔE summaries for the architecture stress cases. These values are not evidence of intrinsic datatype superiority.

Family	Cells	Mean ΔE	Minimum	Maximum
RT-DETR-L	36	-6.79	-54.33	+1.51
RetinaNet-R50-FPN-v2	36	-6.23	-37.75	+1.16

FP8 remained within 1.15 matched-clean AP points of the FP32-typed engine in all six dataset–family blocks. By contrast, the transferred INT8 recipe produced mean clean gaps of 38.66 AP for RT-DETR-L and 22.53 for RetinaNet. Mean corrupted INT8 AP was 1.97 and 11.34, respectively. The supported conclusion is therefore a failure of recipe portability under these recorded treatments, not an intrinsic limitation of INT8 or a universal advantage of FP8.

S13 Evidence-generation and audit boundary

The source package separates three layers:

- **Metric ledgers and evidence reports** contain the retained summaries and upstream SHA-256 bindings supplied with the study.
- **Generation scripts** deterministically reconstruct the decision-impact diagnostic and the canonical cross-family sign convention from packaged CSV inputs.
- **LaTeX** consumes prevalidated tables and figures; compilation itself does not recompute model inference or independently verify external files that are absent from the source package.

The external research pipeline validated metric-to-run, input-manifest, ordered-image, prediction, and analysis hashes before generating the packaged artifacts. The manuscript does not claim that the LaTeX engine performs those checks. The included validator checks package-level consistency: row counts, formula signs, required files, abstract/highlight constraints, graphical abstract dimensions, unresolved placeholders, and SHA-256 manifest integrity. Raw datasets, checkpoints, TensorRT engines, and per-image predictions are not redistributed in this submission source.

S14 Minimum reporting checklist for paired corruption studies

For reuse of this protocol, a report should state at least:

- the exact clean materialization and whether corrupted arms share its terminal codec;
- the treatment-level quantization recipe, graph coverage, calibration identities, runtime version, and decoder;
- the atomic image IDs and proof that compared arms consume identical ordered inputs;
- the direct clean-adjusted contrast, not only corrupted treatment means;
- absolute corrupted AP and explicit low-accuracy floor inventories;
- the resampling unit, pairing structure, draw reuse, and uncertainty boundary;
- the finite-grid weighting target and all non-pooled endpoint families;
- sensitivity to split reuse, corruption realization, training and calibration seeds, class universe, and recipe transfer where feasible.